Consensus Is Not Completion: False Convergence in Multi-Agent Discovery Workflows

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Abstract

Multi-agent language-model workflows often treat agreement, overlap, or a final summary as evidence that a task is complete. This is unsafe in closed-world discovery tasks, where the goal is to find all relevant items. We introduce *false convergence*: a workflow state in which agents report completion and exhibit high agreement, yet the final answer omits true items. We identify two mechanisms: *aggregation loss*, where minority-discovered true items are discarded during consensus or summarization, and *common blind spots*, where agents agree because they miss the same region of the search space. On a real Click repository audit, a standard LLM summarizer self-reports completion in all three seeds while retaining only 68.5–74.5% of oracle items. We argue that completion should be modeled as coverage-risk assessment rather than vote-counting, and propose a lightweight completion certificate that returns SAFE-TO-STOP, UNSAFE-TO-STOP, or REQUIRES-AUDIT from observable workflow signals. When risk is detected, an evidence-preserving audit policy retains singleton evidence and triggers holdout or source-focused review. Experiments on synthetic tasks and two real-repository audits show that the certificate identifies several high-risk false-convergence states, while also exposing the limits of current audit policies.

1 Introduction

Multi-agent language-model systems are increasingly used for tasks that require search, decomposition, and synthesis. A common design pattern is simple: multiple agents independently inspect a task, their outputs are aggregated by majority vote or a summarizer, and the system stops when agents agree or report high confidence. This pattern is natural. Agreement is easy to measure, and summaries provide a clean final answer.

However, agreement is not the same as completion. In closed-world discovery tasks, such as finding every affected file location, every policy case, or every relevant document in a bounded corpus, the central question is not only whether the final answer is plausible. It is whether there remains missing mass outside the discovered set. A group of agents can converge for the wrong reason: they may share a search strategy, omit the same boundary cases, or discard minority evidence during aggregation.

This paper studies this failure mode. We call it *false convergence*: a workflow state in which the system appears complete under its own completion signals but is incomplete under a closed-world oracle. The failure is subtle because the final output can have perfect precision and high self-reported confidence. The error is not necessarily hallucination. It is often omission.

We make three contributions.

- We define false convergence in closed-world agentic discovery and distinguish two mechanisms: aggregation loss and common blind spots.
- We model completion as coverage-risk assessment rather than vote-counting over agent outputs.
- We propose a lightweight completion certificate and an evidence-preserving audit policy, and evaluate them on synthetic tasks and two real-repository audits.

Our results do not imply that union aggregation is always better than consensus. Instead, they show that singleton evidence has two faces. Sometimes it is missing mass; sometimes it is noise. A reliable completion controller should model this distinction rather than silently deleting minority evidence or blindly accepting all unique items.

2 Related Work

2.1 Multi-Agent Aggregation and LLM Agents

Self-consistency selects a consistent answer from multiple samples (Wang et al. 2022); debate, collaboration, and role-specialized agent frameworks improve reasoning or task performance (Du et al. 2023; Chan et al. 2023; Wu et al. 2023; Chen et al. 2023; Hong et al. 2023; Li et al. 2023a; Zhang et al. 2024). Tool-using and planning agents extend LLMs through environment interaction (Yao et al. 2023b; Schick et al. 2023; Qin et al. 2023; Yao et al. 2023a; Wang et al. 2023). Benchmarks such as AgentBench, WebArena, GAIA, and SWE-bench stress interactive or software tasks (Liu et al. 2023; Zhou et al. 2023; Mialon et al. 2023; Jimenez et al. 2023). Our focus is different: aggregation is judged by closed-world recall, not by the quality of one selected answer.

2.2 Verification, Retrieval, and Total Recall

Self-Refine, Reflexion, CRITIC, Chain-of-Verification, and related factuality checks repair or verify outputs (Madaan et al. 2023; Shinn et al. 2023; Gou et al. 2023; Dhuliawala et al. 2023; Lin, Hilton, and Evans 2021; Manakul, Liusie, and Gales 2023; Li et al. 2023b). Retrieval-augmented generation and dense retrieval reduce hallucination but still face noise, negative rejection, integration, and long-context failures (Lewis et al. 2020; Guu et al. 2020; Karpukhin et al. 2020; Nakano et al. 2021; Chen et al. 2024; Li et al. 2024; Liu et al. 2024; Asai et al. 2023; Gao et al. 2023). Closed-world discovery is also related to high-recall retrieval, fact verification, and technology-assisted review, where stopping is a coverage-estimation problem (Thorne et al. 2018; Yang et al. 2018; Grossman, Cormack, and Roegiest 2015; Cormack and Grossman 2021). Our certificate adopts this stopping perspective for correlated LLM-agent exploration.

2.3 Uncertainty and Coverage Estimation

Our certificate uses observable missing-mass signals rather than oracle labels at decision time. Singleton counts and unseen-mass estimates are conceptually related to Good-Turing, Chao-style, capture-recapture, and ecological diversity estimators (Good 1953; Chao 1984; McAllester and Schapire 2000; Chao et al. 2014). Recent LLM uncertainty methods estimate answer risk from probabilities, samples, or semantic variation (Kuhn, Gal, and Farquhar 2023; Kadavath et al. 2022; Cole et al. 2023). We study a complementary question: whether a multi-agent workflow has covered a bounded item set well enough to stop.

3 Problem Formulation

3.1 Closed-World Discovery Tasks

We consider tasks with a bounded source collection D and a finite oracle set $Y^\star$ of target items. A run returns a set $\hat{Y}$. We evaluate:

$$\text{Recall}(\hat{Y}) = \frac{|\hat{Y} \cap Y^\star|}{|Y^\star|}, \tag{1}$$

$$\text{Precision}(\hat{Y}) = \frac{|\hat{Y} \cap Y^\star|}{|\hat{Y}|}. \tag{2}$$

Examples include code dependency audits, policy-case discovery, high-recall document review, and bounded repository analysis. Unlike open-ended question answering, a plausible final answer is insufficient: the system must decide whether it has found the complete set.

3.2 Multi-Agent Completion Signals

A typical workflow runs k agents $A_1, \ldots, A_k$, each producing an item set Y_i, a completion flag c_i, and a confidence score q_i. Aggregation then produces a final set Y_{agg}. Common aggregation rules include:

- **Majority consensus**: keep items reported by at least two agents.
- **Standard summarization**: ask a summarizer to produce a clean final answer from the agent reports.
- **Raw union**: keep every unique item reported by any agent.
- **Union-preserving summarization**: instruct a summarizer to preserve all unique reported items unless malformed or contradicted.

Systems often treat high pairwise overlap, high mean confidence, and affirmative self-reported completion as stopping evidence. We argue that these are not completion certificates.

3.3 False Convergence

Let S be a multi-agent workflow state with G3 agents, an aggregation rule, and an optional holdout scout. We say S exhibits false convergence under an aggregation rule g when:

1. agents self-report completion with high confidence;
2. agent outputs exhibit high overlap;
3. the aggregated answer has recall below a target threshold θ;
4. an independent holdout or oracle audit can recover missing true items.

Our preregistered scoring uses $\theta = 0.95$, confidence threshold 0.80, and overlap threshold $\tau = 0.70$. For strict positive labeling with holdout, we also require holdout gain at least $\gamma = \max(0.05, 3/|Y^\star|)$.

The strict label is intentionally conservative. We also report near-positive and mechanism-positive cases when the missing items are few but diagnostically important.

3.4 Aggregation Loss

Aggregation loss occurs when a true item appears in at least one agent report but is removed by consensus-style aggregation or standard summarization. This is not a search failure: the evidence exists in the reports. The failure happens at the final synthesis step.

Formally, an oracle item $y \in Y^\star$ is an aggregation-loss item if:

$$y \in \bigcup_i Y_i \quad \text{and} \quad y \notin Y_{\text{agg}}. \tag{3}$$

When y appears in only one agent report, it is a singleton true positive.

3.5 Common Blind Spots

Common blind spots occur when multiple agents agree because they share a search failure. In this case, the missing item is absent even from the raw union:

$$y \in Y^\star \quad \text{and} \quad y \notin \bigcup_i Y_i. \tag{4}$$

Raw union and union-preserving summarization cannot recover such items because there is no reported evidence to preserve. The mitigation must change the search or auditing process, for example through boundary-focused holdout review.

3.6 Singletons as Missing Mass or Noise

Singletons are ambiguous. A singleton may be a hard-to-find true positive, or it may be a false positive. Consensus implicitly treats singletons as noise. Raw union implicitly treats them as valid evidence. Both assumptions are unsafe. The right action is often to preserve singleton evidence for audit.

4 Coverage-Risk Completion Certificate

We propose a lightweight completion certificate for deciding whether a multi-agent discovery workflow can safely stop. The certificate is not an oracle: it does not use ground truth and does not decide final item labels. Instead, it maps observable workflow signals to a stopping judgment.

4.1 Observable Signals

The certificate takes as input the G3 agent reports, optional holdout reports, and metadata produced during the run. We use the following observable signals:

- **Confidence**: mean self-reported confidence and completion flags from the agents or summarizer.
- **Overlap**: pairwise Jaccard overlap among agent item sets.
- **Singleton evidence**: items reported by exactly one agent, which may indicate either missing mass or noise.
- **Holdout gain**: new verified items contributed by an independent holdout scout when available.
- **Unseen-mass estimate**: a Chao-style estimate derived from singleton and low-frequency item counts.
- **Task-risk triggers**: whether the task has boundary-resolution risk, such as indirect references, adapter registries, changelog boundaries, or source regions known to be easy to skip.

These signals are intentionally lightweight. They do not prove completion; they indicate whether the current workflow state has enough coverage risk that stopping would be unsafe.

4.2 Certificate Outputs

The certificate returns one of three labels:

- SAFE-TO-STOP: observable risk is low enough that the workflow may publish the current conservative answer.
- UNSAFE-TO-STOP: observable risk suggests likely missing mass; the workflow should continue searching or run a stronger audit.
- REQUIRES-AUDIT: the current answer may be complete, but singleton evidence, noisy union items, or weak independence prevents clean certification.

In the current experiments, the certificate is deliberately conservative. It is designed to catch high-risk states rather than to prove that all safe states are complete. We therefore report it as a coverage-risk certificate, not as a general solution to stopping in all agent workflows.

4.3 Evidence-Preserving Audit Policy

When the certificate returns UNSAFE-TO-STOP or REQUIRES-AUDIT, the workflow needs an audit policy. We use an evidence-preserving policy that separates finalization from evidence retention: consensus items form a conservative answer, singleton evidence is retained for review, and high-agreement boundary-risk states trigger holdout or source-focused audit. This policy is the audit response to a risky certificate state, not the certificate itself.

Algorithm 1: Evidence-Preserving Audit Policy

Run k independent discovery agents, producing $Y_1, \ldots, Y_k$.
Compute item counts $n(y) = |\{i : y \in Y_i\}|$.
Set conservative final set $F = \{y : n(y) \geq 2\}$.
Set audit queue $Q = \{y : n(y) = 1\}$.
Compute confidence, overlap, holdout gain, and unseen-mass signals.
Apply the coverage-risk completion certificate.
if certificate is SAFE-TO-STOP **then**
    Return F with certificate report.
end if
if Q is non-empty **then**
    Trigger singleton audit.
end if
if confidence and overlap are high and the task has boundary-resolution risk **then**
    Trigger boundary-focused holdout audit.
end if
Add audit-verified singletons to F.
Add holdout-discovered boundary items to F.
Return F plus a coverage-risk report.

The audit policy handles the two failure mechanisms differently. For aggregation loss, it preserves and audits singleton or minority evidence instead of deleting it during summarization. For common blind spots, where no current agent reported the missing item, it triggers boundary-focused holdout or source-focused review. In our experiments, singletons are not automatically accepted; they are retained and audited. High agreement is not automatically trusted; it can trigger a common-blind-spot check.

5 Experimental Setup

5.1 Research Questions

We ask:

- **RQ1**: Can consensus-style aggregation produce false completion signals in closed-world discovery tasks?
- **RQ2**: Are failures caused by aggregation loss, common blind spots, or both?
- **RQ3**: Does raw union solve the problem, or does it introduce a precision cost?
- **RQ4**: Can a coverage-risk certificate identify unsafe stopping states, and when can an audit policy recover missing items?

5.2 Tasks

We use four bounded discovery task families: two synthetic families designed to isolate mechanisms, and two real-repository audits used to test whether the failure appears beyond constructed examples.

T1: Code dependency audit. Agents inspect a synthetic AcmePay migration repository and must find every code or configuration location that resolves to AcmePay v1 through direct calls, routing configuration, registry indirection, fallback queues, or replay scripts. The hard oracle has 35 items.

T2: Policy document search. Agents inspect naturalistic incident notes, changelogs, appendices, and policy memos. They must find all cases whose state, flow, and service-resolution chain imply AcmePay v1 handling. The oracle has 30 items. Boundary cases require following service catalog and adapter registry resolution rather than relying on surface service names.

T3: Real-repository Click deprecation audit. Agents inspect a fixed snapshot of the public Click repository and must find line-level locations that belong to a deprecated API surface audit. The oracle has 149 items spanning implementation, warning emission, help-message formatting, parameter behavior, tests, documentation, and bounded changelog entries. Unlike T1/T2, T3 uses a real repository and is used as a stronger validation of aggregation behavior rather than as a strict false-convergence construction.

T4: Real-repository Requests TLS audit. We additionally instantiate a second real-repository task family from a fixed snapshot of the public Requests repository. The task asks agents to find line-level items related to TLS certificate verification, CA bundle resolution, client certificates, SSL error handling, tests, test infrastructure, and user documentation. The oracle has 304 items. T4 is used to test whether the real-repository result depends on Click or deprecation-surface search.

5.3 Runs and Aggregation Policies

Each seed contains three G3 agents. Where available, we also run an independent G6 holdout scout. We compare:

- majority consensus;
- blind standard summarization;
- raw union;
- blind union-preserving summarization;
- holdout scout;
- the coverage-risk completion certificate with evidence-preserving audit policy.

The standard and union-preserving summarizers are blind to oracle and scoring files. The certificate and audit policy use only G3 outputs, optional holdout outputs, confidence, overlap, and task-risk triggers; oracle is used only for evaluation.

6 Results

6.1 Mechanism Evidence

Table 1 summarizes the six current case slices. T1 seed01 and T2 seed01 show aggregation loss: true items were present in minority reports but dropped by consensus and standard summaries. T2 seed02 and seed03 show common blind spots: all three G3 agents agreed on the same incomplete 28-item set, and raw union could not recover the missing cases. T1 seed02 and seed03 are precision-cost controls: consensus is complete, while raw union preserves false positive singletons.

6.2 Aggregation Policy Comparison

Table 2 shows the method-level comparison. Consensus and standard summarization can produce high-precision false stops. Raw union and union-preserving summarization recover aggregation-loss cases, but incur precision cost in T1 seed02/03 and cannot recover T2 common blind spots. The evidence-preserving audit policy recovers missing items when holdout evidence is available and marks singleton-noise controls as requiring audit rather than silently accepting false positives.

6.3 Real-Repository Summarizer False Stops

T3 evaluates whether the aggregation-stage failure appears in a real repository task rather than only in constructed mechanism slices. We give the same G3 aggregation packet to two blind LLM summarizers. The *standard* summarizer is instructed to produce a concise, high-precision final answer from the three agent reports. The *union-preserving* summarizer is instructed to preserve all unique reported items unless malformed or contradicted.

Table 3 shows a stable pattern across three blind seeds. The standard summarizer reports completion with confidence between 0.93 and 0.96, but its recall is only 0.685–0.745. Its precision is nearly perfect, so the failure is not hallucination; it is omission. The union-preserving summarizer removes the false stop in all three seeds, reaching 0.993–1.000 recall, but precision falls to 0.639–0.744. This supports two claims. First, much of the missing evidence was already present in the agent reports and was discarded during aggregation. Second, naive union preservation is diagnostic but not a complete solution because it preserves noise along with missing true items.

6.4 Second Real-Repository Task Family

T4 evaluates a different real repository and audit surface: TLS and certificate handling in Requests. Unlike T3, this task is larger and noisier. Table 4 shows that the failure shifts from mostly aggregation loss to mixed search-coverage and aggregation risk. Majority consensus is incomplete in all three seeds, with recall between 0.678 and 0.701. Raw union improves recall to 0.829–0.859, showing that some omitted evidence is present in minority reports, but raw union remains below the 0.95 completion threshold in every seed. Thus T4 contains candidate-pool missing mass that union aggregation cannot recover.

The standard summarizer also false-stops in all three seeds, with recall 0.368–0.648 despite self-reported completion. The completion certificate marks all three seeds UNSAFE-TO-STOP. This supports the broader claim that a completion controller must estimate coverage risk, not

Case	Mechanism	Mean Conf.	Mean Jac.	Singletons	Cons. Miss.	Union FP	Strict
T1 hard seed01	aggregation loss	0.853	0.752	13	13	0	yes
T1 hard seed02	precision control	0.907	0.932	4	0	4	no
T1 hard seed03	precision control	0.893	0.685	5	0	5	no
T2 docs seed01	aggregation loss	0.887	0.956	2	2	0	no
T2 docs seed02	common blind spot	0.937	1.000	0	2	0	no
T2 docs seed03	common blind spot	0.920	1.000	0	2	0	no

Table 1: Mechanism-level evidence. T2 seeds are not strict positives because holdout gain is $2/30 = 0.067 < \gamma = 0.100$, but seed02/03 repeatedly show high agreement with the same missing boundary cases.

Case	Method	Found	TP	FP	Recall	Status
T1 seed01	consensus / standard	22	22	0	0.629	false stop
T1 seed01	raw union / union-preserving	35	35	0	1.000	complete
T1 seed01	audit policy	35	35	0	1.000	verified
T1 seed02	consensus / standard	35	35	0	1.000	complete
T1 seed02	raw union / union-preserving	39	35	4	1.000	precision cost
T1 seed02	audit policy	35	35	0	1.000	requires audit
T1 seed03	consensus / standard	35	35	0	1.000	complete
T1 seed03	raw union / union-preserving	40	35	5	1.000	precision cost
T1 seed03	audit policy	35	35	0	1.000	requires audit
T2 seed01	consensus / standard	28	28	0	0.933	false stop
T2 seed01	raw union / union-preserving	30	30	0	1.000	complete
T2 seed01	audit policy	30	30	0	1.000	verified
T2 seed02	consensus / raw union	28	28	0	0.933	common blind spot
T2 seed02	holdout / audit policy	30	30	0	1.000	verified
T2 seed03	consensus / raw union	28	28	0	0.933	common blind spot
T2 seed03	holdout / audit policy	30	30	0	1.000	verified

Table 2: Method comparison on current case slices. Grouped rows combine methods with identical scores in the current experiments.

merely summarize or vote over the current reports. T4 should not be read as a direct copy of the T3 mechanism; instead, it is a harder boundary case showing that real-repository discovery can combine aggregation loss, noisy singleton evidence, and insufficient search coverage.

6.5 Completion Certificate on T3

The real-repository results also reveal a limitation of audit-by-holdout alone. In T3 seed01, the evidence-preserving audit policy reaches 0.993 recall and 0.955 precision by auditing singleton evidence. In seed02 and seed03, however, the available holdout scout is itself incomplete, so the audit policy recovers only to 0.805 and 0.832 recall. The completion certificate catches this boundary: for all three T3 seeds it returns UNSAFE-TO-STOP, triggered by low mean confidence, singleton missing-mass signals, and Chao unseen-mass estimates. Thus the method contribution should be read in two layers: evidence preservation can recover aggregation-stage omissions when audit evidence is strong, while the certificate prevents high-risk states from being falsely certified as complete.

6.6 Source-Aware Audit Prototype

The T3 certificate tells the workflow not to stop, but a practical system also needs an audit policy that converts retained evidence into recovery. We evaluate an offline source-aware audit prototype over G3/holdout candidates, using deterministic task-policy predicates over source lines. This is an audit-policy upper bound, not a blind LLM result. On T3, the raw candidate pools average 0.998 recall but only 0.660 precision. Source-aware filtering keeps the same average recall and raises precision to 1.000; a bounded source sweep reaches 1.000 recall and 1.000 precision. On T4, candidate filtering reaches 0.849 recall and 1.000 precision, showing that precise audit is possible but search coverage remains insufficient.

6.7 Audit Policy Behavior

The audit policy exposes rather than hides uncertainty. In T1 seed01, it audits and verifies 13 singleton true positives. In T2 seed01, it verifies 2 singleton true positives. In T2 seed02/03, there are no singletons, but high agreement and high confidence trigger boundary-focused holdout, recovering 2 new items in each seed. In T1 seed02/03, the audit policy leaves 4 and 5 singleton items in the audit queue and

Seed	Policy	Conf.	Found	TP	FP	Recall	Precision	False Stop
01	standard summarizer	0.930	105	104	1	0.698	0.990	yes
01	union-preserving summarizer	0.910	199	148	51	0.993	0.744	no
02	standard summarizer	0.960	111	111	0	0.745	1.000	yes
02	union-preserving summarizer	0.950	233	149	84	1.000	0.639	no
03	standard summarizer	0.930	102	102	0	0.685	1.000	yes
03	union-preserving summarizer	0.840	232	149	83	1.000	0.642	no
Mean	standard summarizer	–	–	–	–	0.709	0.997	3/3
Mean	union-preserving summarizer	–	–	–	–	0.998	0.675	0/3

Table 3: T3 real-repository summarizer comparison. The oracle contains 149 line-level Click deprecation-audit items. Standard summarization is high-precision but repeatedly false-stops; union-preserving summarization recovers recall but incurs a substantial precision cost.

Seed	Cons. Rec.	Std. Rec.	Union Rec.	Union Prec.	Holdout Rec.	Cert.
01	0.701	0.368	0.859	0.661	0.750	unsafe
02	0.678	0.622	0.845	0.668	0.688	unsafe
03	0.678	0.648	0.829	0.676	0.763	unsafe
Mean	0.685	0.546	0.844	0.668	0.734	3/3 unsafe

Table 4: T4 Requests TLS/certificate audit on a second real repository. Raw union improves over consensus but remains incomplete, indicating missing mass outside the G3 candidate pool. Standard summarization false-stops in all three seeds.

does not mark completion, avoiding raw union's false positives.

6.8 Ablation and Cost Proxy

We additionally run audit-policy ablations. Removing singleton audit leaves the aggregation-loss singletons in T1 seed01 and T2 seed01 unrecovered. Removing the common-blind-spot trigger leaves T2 seed02/03 at 28/30 recall despite perfect G3 agreement. Removing holdout converts hidden risk into a REQUIRES-AUDIT state but cannot recover missing items. Thus the two triggers correspond to distinct failure mechanisms, and holdout is the mechanism that turns retained risk into verified recovery.

Current logs do not contain reliable token or wall-clock accounting, so we report a proxy cost: audit queue size plus one unit per triggered holdout. Under this proxy, T1 seed01 uses 14 audit actions to recover 13 true positives, T2 seed01 uses 3 actions to recover 2 true positives, and T2 seed02/03 use one holdout unit to recover 2 true positives each. In T1 seed02/03, the audit policy avoids the 4 and 5 false positives introduced by raw union, but correctly refuses to certify completion without further audit.

7 Discussion and Limitations

7.1 Why Consensus Fails

Consensus can fail for two opposite reasons. It can be too conservative when a minority report contains real evidence. It can also be overconfident when agents are correlated and share a blind spot. These are not contradictions. They show that agreement is evidence of shared sampling behavior, not necessarily evidence of low missing mass.

7.2 Why Raw Union Is Insufficient

Raw union is a useful diagnostic: if raw union recovers missing items, then the failure occurred after discovery, at aggregation. But raw union is not a general solution. It can preserve singleton false positives, and it cannot recover items that no agent reported. A production workflow needs an audit policy, not merely a larger set operation.

7.3 Completion as Coverage-Risk Assessment

The core lesson is that completion should be represented as a risk assessment. A final answer should be accompanied by signals such as singleton count, singleton audit status, pairwise overlap, source coverage, holdout gain, and known boundary-risk triggers. This reframes agent completion from a consensus problem to an uncertainty-estimation problem under correlated exploration.

7.4 Limitations and Next Experiments

The evidence supports mechanism-level claims, real-repository false stops, and a proof-of-concept coverage-risk certificate, not a general proof of safe completion. Broader claims require more repositories, task families, seeds, agent counts, prompt-diverse and model-heterogeneous runs, blind source-partitioned audits, systematic token and wall-clock accounting, stronger stopping baselines, and correlation metrics beyond output Jaccard.

8 Conclusion

Consensus is not completion. In closed-world discovery tasks, multi-agent agreement and high confidence can hide missing true items. We identified two mechanisms: aggregation loss, where true singleton evidence is discarded, and common blind spots, where agents agree because they miss the same boundary cases. We showed that raw union diagnoses some aggregation failures but can preserve false positives and cannot recover unreported items. A coverage-risk certificate separates the stopping decision from the audit policy: it can mark high-risk states as unsafe or requiring audit, while evidence-preserving audit retains minority evidence and triggers additional review. These results suggest that reliable agentic workflows need completion certificates grounded in coverage risk, not only consensus.

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